

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
11.1	Scope	N/A	Functional	No Relationship	N/A	N/A	N/A	0	
11.1(a)	Scope	The regulations in this part set forth the criteria under which the agency considers electronic records, electronic signatures, and handwritten signatures executed to electronic records to be trustworthy, reliable, and generally equivalent to paper records and handwritten signatures executed on paper.	Functional	No Relationship	N/A	N/A	N/A	0	
11.1(b)	Scope	This part applies to records in electronic form that are created, modified, maintained, archived, retrieved, or transmitted, under any records requirements set forth in agency regulations. This part also applies to electronic records submitted to the agency under requirements of the Federal Food, Drug, and Cosmetic Act and the Public Health Service Act, even if such records are not specifically identified in agency regulations. However, this part does not apply to paper records that are, or have been, transmitted by electronic means.	Functional	No Relationship	N/A	N/A	N/A	0	
11.1(c)	Scope	Where electronic signatures and their associated electronic records meet the requirements of this part, the agency will consider the electronic signatures to be equivalent to full handwritten signatures, initials, and other general signings as required by agency regulations, unless specifically excepted by regulation(s) effective on or after August 20, 1997.	Functional	No Relationship	N/A	N/A	N/A	0	
11.1(d)	Scope	Electronic records that meet the requirements of this part may be used in lieu of paper records, in accordance with §11.2, unless paper records are specifically required.	Functional	No Relationship	N/A	N/A	N/A	0	
11.1(e)	Scope	Computer systems (including hardware and software), controls, and attendant documentation maintained under this part shall be readily available for, and subject to, FDA inspection.	Functional	No Relationship	N/A	N/A	N/A	0	
11.1(f)	Scope	This part does not apply to records required to be established or maintained by §§1.326 through 1.368 of this chapter. Records that satisfy the requirements of part 1, subpart J of this chapter, but that also are required under other applicable statutory provisions or regulations, remain subject to this part.	Functional	No Relationship	N/A	N/A	N/A	0	
11.1(g)	Scope	This part does not apply to electronic signatures obtained under §101.11(d) of this chapter.	Functional	No Relationship	N/A	N/A	N/A	0	
11.1(h)	Scope	This part does not apply to electronic signatures obtained under §101.8(d) of this chapter.	Functional	No Relationship	N/A	N/A	N/A	0	
11.1(i)	Scope	This part does not apply to records required to be established or maintained by part 117 of this chapter. Records that satisfy the requirements of part 117 of this chapter, but that also are required under other applicable statutory provisions or regulations, remain subject to this part.	Functional	No Relationship	N/A	N/A	N/A	0	
11.1(j)	Scope	This part does not apply to records required to be established or maintained by part 507 of this chapter. Records that satisfy the requirements of part 507 of this chapter, but that also are required under other applicable statutory provisions or regulations, remain subject to this part.	Functional	No Relationship	N/A	N/A	N/A	0	
11.1(k)	Scope	This part does not apply to records required to be established or maintained by part 112 of this chapter. Records that satisfy the requirements of part 112 of this chapter, but that also are required under other applicable statutory provisions or regulations, remain subject to this part.	Functional	No Relationship	N/A	N/A	N/A	0	
11.1(l)	Scope	This part does not apply to records required to be established or maintained by subpart L of part 1 of this chapter. Records that satisfy the requirements of subpart L of part 1 of this chapter, but that also are required under other applicable statutory provisions or regulations, remain subject to this part.	Functional	No Relationship	N/A	N/A	N/A	0	
11.1(m)	Scope	This part does not apply to records required to be established or maintained by subpart M of part 1 of this chapter. Records that satisfy the requirements of subpart M of part 1 of this chapter, but that also are required under other applicable statutory provisions or regulations, remain subject to this part.	Functional	No Relationship	N/A	N/A	N/A	0	
11.1(n)	Scope	This part does not apply to records required to be established or maintained by subpart O of part 1 of this chapter. Records that satisfy the requirements of subpart O of part 1 of this chapter, but that also are required under other applicable statutory provisions or regulations, remain subject to this part.	Functional	No Relationship	N/A	N/A	N/A	0	
11.1(o)	Scope	This part does not apply to records required to be established or maintained by part 121 of this chapter. Records that satisfy the requirements of part 121 of this chapter, but that also are required under other applicable statutory provisions or regulations, remain subject to this part.	Functional	No Relationship	N/A	N/A	N/A	0	
11.1(p)	Scope	This part does not apply to records required to be established or maintained by subpart R of part 1 of this chapter. Records that satisfy the requirements of subpart R of part 1 of this chapter, but that also are required under other applicable statutory provisions or regulations, remain subject to this part.	Functional	No Relationship	N/A	N/A	N/A	0	
11.2	Implementation	N/A	Functional	No Relationship	N/A	N/A	N/A	0	
11.2(a)	Implementation	For records required to be maintained but not submitted to the agency, persons may use electronic records in lieu of paper records or electronic signatures in lieu of traditional signatures, in whole or in part, provided that the requirements of this part are met.	Functional	No Relationship	N/A	N/A	N/A	0	
11.2(b)	Implementation	For records submitted to the agency, persons may use electronic records in lieu of paper records or electronic signatures in lieu of traditional signatures, in whole or in part, provided that:	Functional	No Relationship	N/A	N/A	N/A	0	
11.2(b)(1)	Implementation	The requirements of this part are met; and	Functional	No Relationship	N/A	N/A	N/A	0	
11.2(b)(2)	Implementation	The document or parts of a document to be submitted have been identified in public docket No. 925-0251 as being the type of submission the agency accepts in electronic form. This docket will identify specifically what types of documents or parts of documents are acceptable for submission in electronic form without paper records and the agency receiving unit(s) (e.g., specific center, office, division, branch) to which such submissions may be made. Documents to agency receiving unit(s) not specified in the public docket will not be considered as official if they are submitted in electronic form; paper forms of such documents will be considered as official and must accompany any electronic records. Persons are expected to consult with the intended agency receiving unit for details on how (e.g., method of transmission, media, file formats, and technical protocols) and whether to proceed with the electronic submission.	Functional	No Relationship	N/A	N/A	N/A	0	
11.3	Definitions	See FDE for details.	Functional	No Relationship	N/A	N/A	N/A	0	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Subset Of	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient capabilities.	10	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Change Management Program	CHG-01	Mechanisms exist to facilitate the implementation of a change management program.	5	

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11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Statutory, Regulatory & Contractual Compliance	CPL-01	Mechanisms exist to facilitate the identification and implementation of relevant statutory, regulatory and contractual controls.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Security, Compliance & Resilience Controls Oversight	CPL-02	Mechanisms exist to provide a security, compliance and resilience controls oversight function that reports to the organization's executive leadership.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Control Conformity Monitoring	CPL-03	Mechanisms exist to validate that Technology Assets, Applications, Services and/or Data (TAASD) conform to the organization's security, compliance and/or resilience policies, standards and other applicable requirements.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Functional Review Of Security, Compliance & Resilience Controls	CPL-03.2	Mechanisms exist to regularly review Technology Assets, Applications and/or Services (TAAS) for adherence to the organization's security, compliance and/or resilience policies and standards.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Continuous Monitoring	MON-01	Mechanisms exist to facilitate the implementation of enterprise-wide monitoring controls.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Analyze and Prioritize Monitoring Requirements	MON-01.16	Mechanisms exist to assess the organization's needs for monitoring and prioritize the monitoring of Technology Assets, Applications and/or Services (TAAS), based on TAAS criticality and the sensitivity of the data it stores, transmits and processes.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Centralized Collection of Security Event Logs	MON-02	Mechanisms exist to utilize a Security Incident Event Manager (SIEM), or similar automated tool, to support the centralized collection of security-related event logs.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Content of Event Logs	MON-03	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) to produce event logs that contain sufficient information to, at a minimum: (1) Establish what type of event occurred; (2) When (date and time) the event occurred; (3) Where the event occurred; (4) The source of the event; (5) The outcome (success or failure) of the event; and (6) The identity of any user/subject associated with the event.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Audit Trails	MON-03.2	Mechanisms exist to link system access to individual users or service accounts.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Time Stamps	MON-07	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) to use an authoritative time source to generate time stamps for event logs.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Synchronization With Authoritative Time Source	MON-07.1	Mechanisms exist to synchronize internal system clocks with an authoritative time source.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Protection of Event Logs	MON-08	Mechanisms exist to protect event logs and audit tools from unauthorized access, modification and deletion.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Use of Cryptographic Controls	CRY-01	Mechanisms exist to facilitate the implementation of cryptographic protections controls using known public standards and trusted cryptographic technologies.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Transmission Confidentiality	CRY-03	Cryptographic mechanisms exist to protect the confidentiality of data being transmitted.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Transmission Integrity	CRY-04	Cryptographic mechanisms exist to protect the integrity of data being transmitted.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Encrypting Data At Rest	CRY-05	Cryptographic mechanisms exist to prevent unauthorized disclosure of data at rest.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Public Key Infrastructure (PKI)	CRY-08	Mechanisms exist to securely implement an internal Public Key Infrastructure (PKI) infrastructure or obtain PKI services from a reputable PKI service provider.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Cryptographic Key Management	CRY-09	Mechanisms exist to facilitate cryptographic key management controls to protect the confidentiality, integrity and availability of keys.	5	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **US - Food & Drug Administration (FDA) 21 CFR Part 11 (2025)**

Focal Document URL: <https://www.govinfo.gov/content/pkg/CFR-2025-title21-vol1/pdf/CFR-2025-title21-vol1-part11.pdf>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-usa-federal-fda-21-cfr-part-11-2025.pdf>

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11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Data Protection	DCH-01	Mechanisms exist to facilitate the implementation of data protection controls.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Media & Data Retention	DCH-18	Mechanisms exist to retain media and data in accordance with applicable statutory, regulatory and contractual obligations.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Competency Requirements for Security-Related Positions	HRS-03.2	Mechanisms exist to ensure that all security-related positions are staffed by qualified individuals who have the necessary skill set.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Rules of Behavior	HRS-05.1	Mechanisms exist to define acceptable and unacceptable rules of behavior for the use of technologies, including consequences for unacceptable behavior.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Identity & Access Management (IAM)	IAC-01	Mechanisms exist to facilitate the implementation of identification and access management controls.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Identification & Authentication for Organizational Users	IAC-02	Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) organizational users and processes acting on behalf of organizational users.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Identification & Authentication for Devices	IAC-04	Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) devices before establishing a connection using bidirectional authentication that is cryptographically- based and replay resistant.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Role-Based Access Control (RBAC)	IAC-08	Mechanisms exist to enforce Role-Based Access Control (RBAC) for Technology Assets, Applications, Services and/or Data (TAASD) to restrict access to individuals assigned specific roles with legitimate business needs.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Least Privilege	IAC-21	Mechanisms exist to utilize the concept of least privilege, allowing only authorized access to processes necessary to accomplish assigned tasks in accordance with organizational business functions.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Information Assurance (IA) Operations	IAO-01	Mechanisms exist to facilitate the implementation of security, compliance and resilience assessment and authorization controls.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Maintenance Operations	MNT-01	Mechanisms exist to develop, disseminate, review & update procedures to facilitate the implementation of maintenance controls across the enterprise.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Secure Engineering Principles	SEA-01	Mechanisms exist to facilitate the implementation of industry-recognized security, compliance and resilience practices in the specification, design, development, implementation and modification of Technology Assets, Applications and/or Services (TAAS).	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Alignment With Enterprise Architecture	SEA-02	Mechanisms exist to develop an enterprise architecture, aligned with industry-recognized leading practices, with consideration for security, compliance and resilience principles that addresses risk to organizational operations, assets, individuals and other organizations.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Defense-In-Depth (DiD) Architecture	SEA-03	Mechanisms exist to implement security functions as a layered structure minimizing interactions between layers of the design and avoiding any dependence by lower layers on the functionality or correctness of higher layers.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Operations Security	OPS-01	Mechanisms exist to facilitate the implementation of operational security controls.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Standardized Operating Procedures (SOP)	OPS-01.1	Mechanisms exist to identify and document Standardized Operating Procedures (SOP), or similar documentation, to enable the proper execution of day-to-day / assigned tasks.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Service Delivery (Business Process Support)	OPS-03	Mechanisms exist to define supporting business processes and implement appropriate governance and service management to ensure appropriate planning, delivery and support of the organization's technology capabilities supporting business functions, workforce, and/or customers based on industry-recognized standards to achieve the specific goals of the process area.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Security, Compliance & Resilience-Minded Workforce	SAT-01	Mechanisms exist to facilitate the implementation of security workforce development and awareness controls.	5	
11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Role-Based Security, Compliance & Resilience Training	SAT-03	Mechanisms exist to provide role-based security, compliance and resilience-related training: (1) Before authorizing access to the system or performing assigned duties; (2) When required by system changes; and (3) Annually thereafter.	5	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **US - Food & Drug Administration (FDA) 21 CFR Part 11 (2025)**

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11.10	Controls for closed systems	Persons who use closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:	Functional	Intersects With	Malformed Input Testing	TDA-09.4	Mechanisms exist to utilize testing methods to ensure Technology Assets, Applications and/or Services (TAAS) continue to operate as intended when subject to invalid or unexpected inputs on its interfaces.	5	
11.10(a)	Controls for closed systems	Validation of systems to ensure accuracy, reliability, consistent intended performance, and the ability to discern invalid or altered records.	Functional	Intersects With	Security, Compliance & Resilience Controls Oversight	CPL-02	Mechanisms exist to provide a security, compliance and resilience controls oversight function that reports to the organization's executive leadership.	8	
11.10(a)	Controls for closed systems	Validation of systems to ensure accuracy, reliability, consistent intended performance, and the ability to discern invalid or altered records.	Functional	Intersects With	Control Conformity Monitoring	CPL-03	Mechanisms exist to validate that Technology Assets, Applications, Services and/or Data (TAASD) conform to the organization's security, compliance and/or resilience policies, standards and other applicable requirements.	8	
11.10(a)	Controls for closed systems	Validation of systems to ensure accuracy, reliability, consistent intended performance, and the ability to discern invalid or altered records.	Functional	Intersects With	Functional Review Of Security, Compliance & Resilience Controls	CPL-03.2	Mechanisms exist to regularly review Technology Assets, Applications and/or Services (TAAS) for adherence to the organization's security, compliance and/or resilience policies and standards.	8	
11.10(a)	Controls for closed systems	Validation of systems to ensure accuracy, reliability, consistent intended performance, and the ability to discern invalid or altered records.	Functional	Intersects With	Data Protection	DCH-01	Mechanisms exist to facilitate the implementation of data protection controls.	5	
11.10(a)	Controls for closed systems	Validation of systems to ensure accuracy, reliability, consistent intended performance, and the ability to discern invalid or altered records.	Functional	Intersects With	Information Assurance (IA) Operations	IAO-01	Mechanisms exist to facilitate the implementation of security, compliance and resilience assessment and authorization controls.	8	
11.10(a)	Controls for closed systems	Validation of systems to ensure accuracy, reliability, consistent intended performance, and the ability to discern invalid or altered records.	Functional	Intersects With	Assessment Boundaries	IAO-01.1	Mechanisms exist to establish the scope of assessments by defining the assessment boundary, according to people, processes and technology that directly or indirectly impact the confidentiality, integrity, availability and safety of the Technology Assets, Applications, Services and/or Data (TAASD) under review.	5	
11.10(a)	Controls for closed systems	Validation of systems to ensure accuracy, reliability, consistent intended performance, and the ability to discern invalid or altered records.	Functional	Intersects With	Assessments	IAO-02	Mechanisms exist to formally assess the security, compliance and resilience controls in Technology Assets, Applications and/or Services (TAAS) through Information Assurance Program (IAP) activities to determine the extent to which the controls are implemented correctly, operating as intended and producing the desired outcome with respect to meeting expected requirements.	8	
11.10(a)	Controls for closed systems	Validation of systems to ensure accuracy, reliability, consistent intended performance, and the ability to discern invalid or altered records.	Functional	Intersects With	File Integrity Monitoring (FIM)	MON-01.7	Mechanisms exist to utilize a File Integrity Monitor (FIM), or similar change-detection technology, on critical Technology Assets, Applications and/or Services (TAAS) to generate alerts for unauthorized modifications.	5	
11.10(b)	Controls for closed systems	The ability to generate accurate and complete copies of records in both human readable and electronic form suitable for inspection, review, and copying by the agency. Persons should contact the agency if there are any questions regarding the ability of the agency to perform such review and copying of the electronic records.	Functional	Intersects With	Statutory, Regulatory & Contractual Compliance	CPL-01	Mechanisms exist to facilitate the identification and implementation of relevant statutory, regulatory and contractual controls.	5	
11.10(b)	Controls for closed systems	The ability to generate accurate and complete copies of records in both human readable and electronic form suitable for inspection, review, and copying by the agency. Persons should contact the agency if there are any questions regarding the ability of the agency to perform such review and copying of the electronic records.	Functional	Intersects With	Security, Compliance & Resilience Controls Oversight	CPL-02	Mechanisms exist to provide a security, compliance and resilience controls oversight function that reports to the organization's executive leadership.	5	
11.10(b)	Controls for closed systems	The ability to generate accurate and complete copies of records in both human readable and electronic form suitable for inspection, review, and copying by the agency. Persons should contact the agency if there are any questions regarding the ability of the agency to perform such review and copying of the electronic records.	Functional	Intersects With	Control Conformity Monitoring	CPL-03	Mechanisms exist to validate that Technology Assets, Applications, Services and/or Data (TAASD) conform to the organization's security, compliance and/or resilience policies, standards and other applicable requirements.	5	
11.10(b)	Controls for closed systems	The ability to generate accurate and complete copies of records in both human readable and electronic form suitable for inspection, review, and copying by the agency. Persons should contact the agency if there are any questions regarding the ability of the agency to perform such review and copying of the electronic records.	Functional	Intersects With	Functional Review Of Security, Compliance & Resilience Controls	CPL-03.2	Mechanisms exist to regularly review Technology Assets, Applications and/or Services (TAAS) for adherence to the organization's security, compliance and/or resilience policies and standards.	5	
11.10(b)	Controls for closed systems	The ability to generate accurate and complete copies of records in both human readable and electronic form suitable for inspection, review, and copying by the agency. Persons should contact the agency if there are any questions regarding the ability of the agency to perform such review and copying of the electronic records.	Functional	Intersects With	Centralized Collection of Security Event Logs	MON-02	Mechanisms exist to utilize a Security Incident Event Manager (SIEM), or similar automated tool, to support the centralized collection of security-related event logs.	5	
11.10(b)	Controls for closed systems	The ability to generate accurate and complete copies of records in both human readable and electronic form suitable for inspection, review, and copying by the agency. Persons should contact the agency if there are any questions regarding the ability of the agency to perform such review and copying of the electronic records.	Functional	Intersects With	Content of Event Logs	MON-03	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) to produce event logs that contain sufficient information to, at a minimum: (1) Establish what type of event occurred; (2) When (date and time) the event occurred; (3) Where the event occurred; (4) The source of the event; (5) The outcome (success or failure) of the event; and (6) The identity of any user/subject associated with the event.	5	
11.10(b)	Controls for closed systems	The ability to generate accurate and complete copies of records in both human readable and electronic form suitable for inspection, review, and copying by the agency. Persons should contact the agency if there are any questions regarding the ability of the agency to perform such review and copying of the electronic records.	Functional	Intersects With	Audit Trails	MON-03.2	Mechanisms exist to link system access to individual users or service accounts.	5	
11.10(b)	Controls for closed systems	The ability to generate accurate and complete copies of records in both human readable and electronic form suitable for inspection, review, and copying by the agency. Persons should contact the agency if there are any questions regarding the ability of the agency to perform such review and copying of the electronic records.	Functional	Intersects With	Time Stamps	MON-07	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) to use an authoritative time source to generate time stamps for event logs.	5	
11.10(b)	Controls for closed systems	The ability to generate accurate and complete copies of records in both human readable and electronic form suitable for inspection, review, and copying by the agency. Persons should contact the agency if there are any questions regarding the ability of the agency to perform such review and copying of the electronic records.	Functional	Intersects With	Synchronization With Authoritative Time Source	MON-07.1	Mechanisms exist to synchronize internal system clocks with an authoritative time source.	5	
11.10(b)	Controls for closed systems	The ability to generate accurate and complete copies of records in both human readable and electronic form suitable for inspection, review, and copying by the agency. Persons should contact the agency if there are any questions regarding the ability of the agency to perform such review and copying of the electronic records.	Functional	Intersects With	Protection of Event Logs	MON-08	Mechanisms exist to protect event logs and audit tools from unauthorized access, modification and deletion.	5	
11.10(b)	Controls for closed systems	The ability to generate accurate and complete copies of records in both human readable and electronic form suitable for inspection, review, and copying by the agency. Persons should contact the agency if there are any questions regarding the ability of the agency to perform such review and copying of the electronic records.	Functional	Intersects With	Data Protection	DCH-01	Mechanisms exist to facilitate the implementation of data protection controls.	5	
11.10(b)	Controls for closed systems	The ability to generate accurate and complete copies of records in both human readable and electronic form suitable for inspection, review, and copying by the agency. Persons should contact the agency if there are any questions regarding the ability of the agency to perform such review and copying of the electronic records.	Functional	Intersects With	Information Assurance (IA) Operations	IAO-01	Mechanisms exist to facilitate the implementation of security, compliance and resilience assessment and authorization controls.	5	
11.10(c)	Controls for closed systems	Protection of records to enable their accurate and ready retrieval throughout the records retention period.	Functional	Intersects With	Statutory, Regulatory & Contractual Compliance	CPL-01	Mechanisms exist to facilitate the identification and implementation of relevant statutory, regulatory and contractual controls.	5	
11.10(c)	Controls for closed systems	Protection of records to enable their accurate and ready retrieval throughout the records retention period.	Functional	Intersects With	Security, Compliance & Resilience Controls Oversight	CPL-02	Mechanisms exist to provide a security, compliance and resilience controls oversight function that reports to the organization's executive leadership.	5	
11.10(c)	Controls for closed systems	Protection of records to enable their accurate and ready retrieval throughout the records retention period.	Functional	Intersects With	Control Conformity Monitoring	CPL-03	Mechanisms exist to validate that Technology Assets, Applications, Services and/or Data (TAASD) conform to the organization's security, compliance and/or resilience policies, standards and other applicable requirements.	5	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **US - Food & Drug Administration (FDA) 21 CFR Part 11 (2025)**

Focal Document URL: <https://www.govinfo.gov/content/pkg/CFR-2025-title21-vol1/pdf/CFR-2025-title21-vol1-part11.pdf>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-usa-federal-fda-21-cfr-part-11-2025.pdf>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
11.10(c)	Controls for closed systems	Protection of records to enable their accurate and ready retrieval throughout the records retention period.	Functional	Intersects With	Functional Review Of Security, Compliance & Resilience Controls	CPL-03.2	Mechanisms exist to regularly review Technology Assets, Applications and/or Services (TAAS) for adherence to the organization's security, compliance and/or resilience policies and standards.	5	
11.10(c)	Controls for closed systems	Protection of records to enable their accurate and ready retrieval throughout the records retention period.	Functional	Intersects With	Centralized Collection of Security Event Logs	MON-02	Mechanisms exist to utilize a Security Incident Event Manager (SIEM), or similar automated tool, to support the centralized collection of security-related event logs.	5	
11.10(c)	Controls for closed systems	Protection of records to enable their accurate and ready retrieval throughout the records retention period.	Functional	Intersects With	Content of Event Logs	MON-03	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) to produce event logs that contain sufficient information to, at a minimum: (1) Establish what type of event occurred; (2) When (date and time) the event occurred; (3) Where the event occurred; (4) The source of the event; (5) The outcome (success or failure) of the event; and (6) The identity of any user/subject associated with the event.	5	
11.10(c)	Controls for closed systems	Protection of records to enable their accurate and ready retrieval throughout the records retention period.	Functional	Intersects With	Audit Trails	MON-03.2	Mechanisms exist to link system access to individual users or service accounts.	5	
11.10(c)	Controls for closed systems	Protection of records to enable their accurate and ready retrieval throughout the records retention period.	Functional	Intersects With	Time Stamps	MON-07	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) to use an authoritative time source to generate time stamps for event logs.	5	
11.10(c)	Controls for closed systems	Protection of records to enable their accurate and ready retrieval throughout the records retention period.	Functional	Intersects With	Synchronization With Authoritative Time Source	MON-07.1	Mechanisms exist to synchronize internal system clocks with an authoritative time source.	5	
11.10(c)	Controls for closed systems	Protection of records to enable their accurate and ready retrieval throughout the records retention period.	Functional	Intersects With	Protection of Event Logs	MON-08	Mechanisms exist to protect event logs and audit tools from unauthorized access, modification and deletion.	5	
11.10(c)	Controls for closed systems	Protection of records to enable their accurate and ready retrieval throughout the records retention period.	Functional	Intersects With	Use of Cryptographic Controls	CRY-01	Mechanisms exist to facilitate the implementation of cryptographic protections controls using known public standards and trusted cryptographic technologies.	5	
11.10(c)	Controls for closed systems	Protection of records to enable their accurate and ready retrieval throughout the records retention period.	Functional	Intersects With	Transmission Confidentiality	CRY-03	Cryptographic mechanisms exist to protect the confidentiality of data being transmitted.	5	
11.10(c)	Controls for closed systems	Protection of records to enable their accurate and ready retrieval throughout the records retention period.	Functional	Intersects With	Transmission Integrity	CRY-04	Cryptographic mechanisms exist to protect the integrity of data being transmitted.	5	
11.10(c)	Controls for closed systems	Protection of records to enable their accurate and ready retrieval throughout the records retention period.	Functional	Intersects With	Encrypting Data At Rest	CRY-05	Cryptographic mechanisms exist to prevent unauthorized disclosure of data at rest.	5	
11.10(c)	Controls for closed systems	Protection of records to enable their accurate and ready retrieval throughout the records retention period.	Functional	Intersects With	Data Protection	DCH-01	Mechanisms exist to facilitate the implementation of data protection controls.	5	
11.10(c)	Controls for closed systems	Protection of records to enable their accurate and ready retrieval throughout the records retention period.	Functional	Intersects With	Information Assurance (IA) Operations	IAO-01	Mechanisms exist to facilitate the implementation of security, compliance and resilience assessment and authorization controls.	5	
11.10(d)	Controls for closed systems	Limiting system access to authorized individuals.	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.	5	
11.10(d)	Controls for closed systems	Limiting system access to authorized individuals.	Functional	Intersects With	Data Protection	DCH-01	Mechanisms exist to facilitate the implementation of data protection controls.	5	
11.10(d)	Controls for closed systems	Limiting system access to authorized individuals.	Functional	Intersects With	Identity & Access Management (IAM)	IAC-01	Mechanisms exist to facilitate the implementation of identification and access management controls.	5	
11.10(d)	Controls for closed systems	Limiting system access to authorized individuals.	Functional	Intersects With	Identification & Authentication for Organizational Users	IAC-02	Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) organizational users and processes acting on behalf of organizational users.	5	
11.10(d)	Controls for closed systems	Limiting system access to authorized individuals.	Functional	Intersects With	Role-Based Access Control (RBAC)	IAC-08	Mechanisms exist to enforce Role-Based Access Control (RBAC) for Technology Assets, Applications, Services and/or Data (TAASD) to restrict access to individuals assigned specific roles with legitimate business needs.	5	
11.10(d)	Controls for closed systems	Limiting system access to authorized individuals.	Functional	Intersects With	Access Enforcement	IAC-20	Mechanisms exist to enforce Logical Access Control (LAC) permissions that conform to the principle of "least privilege."	5	
11.10(d)	Controls for closed systems	Limiting system access to authorized individuals.	Functional	Intersects With	Least Privilege	IAC-21	Mechanisms exist to utilize the concept of least privilege, allowing only authorized access to processes necessary to accomplish assigned tasks in accordance with organizational business functions.	5	
11.10(e)	Controls for closed systems	Use of secure, computer-generated, time-stamped audit trails to independently record the date and time of operator entries and actions that create, modify, or delete electronic records. Record changes shall not obscure previously recorded information. Such audit trail documentation shall be retained for a period at least as long as that required for the subject electronic records and shall be available for agency review and copying.	Functional	Intersects With	Media & Data Retention	DCH-18	Mechanisms exist to retain media and data in accordance with applicable statutory, regulatory and contractual obligations.	8	
11.10(e)	Controls for closed systems	Use of secure, computer-generated, time-stamped audit trails to independently record the date and time of operator entries and actions that create, modify, or delete electronic records. Record changes shall not obscure previously recorded information. Such audit trail documentation shall be retained for a period at least as long as that required for the subject electronic records and shall be available for agency review and copying.	Functional	Intersects With	Continuous Monitoring	MON-01	Mechanisms exist to facilitate the implementation of enterprise-wide monitoring controls.	8	
11.10(e)	Controls for closed systems	Use of secure, computer-generated, time-stamped audit trails to independently record the date and time of operator entries and actions that create, modify, or delete electronic records. Record changes shall not obscure previously recorded information. Such audit trail documentation shall be retained for a period at least as long as that required for the subject electronic records and shall be available for agency review and copying.	Functional	Intersects With	Centralized Collection of Security Event Logs	MON-02	Mechanisms exist to utilize a Security Incident Event Manager (SIEM), or similar automated tool, to support the centralized collection of security-related event logs.	8	
11.10(e)	Controls for closed systems	Use of secure, computer-generated, time-stamped audit trails to independently record the date and time of operator entries and actions that create, modify, or delete electronic records. Record changes shall not obscure previously recorded information. Such audit trail documentation shall be retained for a period at least as long as that required for the subject electronic records and shall be available for agency review and copying.	Functional	Intersects With	Content of Event Logs	MON-03	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) to produce event logs that contain sufficient information to, at a minimum: (1) Establish what type of event occurred; (2) When (date and time) the event occurred; (3) Where the event occurred; (4) The source of the event; (5) The outcome (success or failure) of the event; and (6) The identity of any user/subject associated with the event.	8	
11.10(e)	Controls for closed systems	Use of secure, computer-generated, time-stamped audit trails to independently record the date and time of operator entries and actions that create, modify, or delete electronic records. Record changes shall not obscure previously recorded information. Such audit trail documentation shall be retained for a period at least as long as that required for the subject electronic records and shall be available for agency review and copying.	Functional	Intersects With	Audit Trails	MON-03.2	Mechanisms exist to link system access to individual users or service accounts.	8	
11.10(e)	Controls for closed systems	Use of secure, computer-generated, time-stamped audit trails to independently record the date and time of operator entries and actions that create, modify, or delete electronic records. Record changes shall not obscure previously recorded information. Such audit trail documentation shall be retained for a period at least as long as that required for the subject electronic records and shall be available for agency review and copying.	Functional	Intersects With	Time Stamps	MON-07	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) to use an authoritative time source to generate time stamps for event logs.	8	
11.10(e)	Controls for closed systems	Use of secure, computer-generated, time-stamped audit trails to independently record the date and time of operator entries and actions that create, modify, or delete electronic records. Record changes shall not obscure previously recorded information. Such audit trail documentation shall be retained for a period at least as long as that required for the subject electronic records and shall be available for agency review and copying.	Functional	Intersects With	Synchronization With Authoritative Time Source	MON-07.1	Mechanisms exist to synchronize internal system clocks with an authoritative time source.	8	
11.10(e)	Controls for closed systems	Use of secure, computer-generated, time-stamped audit trails to independently record the date and time of operator entries and actions that create, modify, or delete electronic records. Record changes shall not obscure previously recorded information. Such audit trail documentation shall be retained for a period at least as long as that required for the subject electronic records and shall be available for agency review and copying.	Functional	Intersects With	Protection of Event Logs	MON-08	Mechanisms exist to protect event logs and audit tools from unauthorized access, modification and deletion.	8	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **US - Food & Drug Administration (FDA) 21 CFR Part 11 (2025)**

Focal Document URL: <https://www.govinfo.gov/content/pkg/CFR-2025-title21-vol1/pdf/CFR-2025-title21-vol1-part11.pdf>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-usa-federal-fda-21-cfr-part-11-2025.pdf>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
11.10(f)	Controls for closed systems	Use of operational system checks to enforce permitted sequencing of steps and events, as appropriate.	Functional	Subset Of	Standardized Operating Procedures (SOP)	OPS-01.1	Mechanisms exist to identify and document Standardized Operating Procedures (SOP), or similar documentation, to enable the proper execution of day-to-day / assigned tasks.	10	
11.10(g)	Controls for closed systems	Use of authority checks to ensure that only authorized individuals can use the system, electronically sign a record, access the operation or computer system input or output device, alter a record, or perform the operation at hand.	Functional	Subset Of	Identity & Access Management (IAM)	IAC-01	Mechanisms exist to facilitate the implementation of identification and access management controls.	10	
11.10(g)	Controls for closed systems	Use of authority checks to ensure that only authorized individuals can use the system, electronically sign a record, access the operation or computer system input or output device, alter a record, or perform the operation at hand.	Functional	Intersects With	Periodic Review of Account Privileges	IAC-17	Mechanisms exist to periodically-review the privileges assigned to individuals and service accounts to validate the need for such privileges and reassign or remove unnecessary privileges, as necessary.	8	
11.10(h)	Controls for closed systems	Use of device (e.g., terminal) checks to determine, as appropriate, the validity of the source of data input or operational instruction.	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	8	
11.10(i)	Controls for closed systems	Determination that persons who develop, maintain, or use electronic record/electronic signature systems have the education, training, and experience to perform their assigned tasks.	Functional	Subset Of	Human Resources Security Management	HRS-01	Mechanisms exist to facilitate the implementation of personnel security controls.	10	
11.10(i)	Controls for closed systems	Determination that persons who develop, maintain, or use electronic record/electronic signature systems have the education, training, and experience to perform their assigned tasks.	Functional	Intersects With	Position Categorization	HRS-02	Mechanisms exist to manage personnel security risk by assigning a risk designation to all positions and establishing screening criteria for individuals filling those positions.	5	
11.10(i)	Controls for closed systems	Determination that persons who develop, maintain, or use electronic record/electronic signature systems have the education, training, and experience to perform their assigned tasks.	Functional	Intersects With	Users With Elevated Privileges	HRS-02.1	Mechanisms exist to ensure that every user accessing Technology Assets, Applications and/or Services (TAAS) that process, store and/or transmit sensitive and/or regulated data is cleared and regularly trained to handle the information in question.	5	
11.10(i)	Controls for closed systems	Determination that persons who develop, maintain, or use electronic record/electronic signature systems have the education, training, and experience to perform their assigned tasks.	Functional	Intersects With	Defined Roles & Responsibilities	HRS-03	Mechanisms exist to define cybersecurity roles & responsibilities for all personnel.	5	
11.10(i)	Controls for closed systems	Determination that persons who develop, maintain, or use electronic record/electronic signature systems have the education, training, and experience to perform their assigned tasks.	Functional	Intersects With	Competency Requirements for Security-Related Positions	HRS-03.2	Mechanisms exist to ensure that all security-related positions are staffed by qualified individuals who have the necessary skill set.	5	
11.10(i)	Controls for closed systems	Determination that persons who develop, maintain, or use electronic record/electronic signature systems have the education, training, and experience to perform their assigned tasks.	Functional	Intersects With	Formal Indoctrination	HRS-04.2	Mechanisms exist to formally educate authorized users on proper data handling practices for all the relevant types of data to which they have access.	5	
11.10(j)	Controls for closed systems	The establishment of, and adherence to, written policies that hold individuals accountable and responsible for actions initiated under their electronic signatures, in order to deter record and signature falsification.	Functional	Subset Of	Human Resources Security Management	HRS-01	Mechanisms exist to facilitate the implementation of personnel security controls.	10	
11.10(j)	Controls for closed systems	The establishment of, and adherence to, written policies that hold individuals accountable and responsible for actions initiated under their electronic signatures, in order to deter record and signature falsification.	Functional	Intersects With	Onboarding, Transferring & Offboarding Personnel	HRS-01.1	Mechanisms exist to proactively govern the following personnel management actions: (1) Onboarding new personnel (e.g., new hires); (2) Transferring personnel into new roles within the organization; and (3) Offboarding personnel (e.g., termination of employment).	5	
11.10(j)	Controls for closed systems	The establishment of, and adherence to, written policies that hold individuals accountable and responsible for actions initiated under their electronic signatures, in order to deter record and signature falsification.	Functional	Intersects With	Formal Indoctrination	HRS-04.2	Mechanisms exist to formally educate authorized users on proper data handling practices for all the relevant types of data to which they have access.	5	
11.10(j)	Controls for closed systems	The establishment of, and adherence to, written policies that hold individuals accountable and responsible for actions initiated under their electronic signatures, in order to deter record and signature falsification.	Functional	Intersects With	Terms of Employment	HRS-05	Mechanisms exist to require all employees and contractors to apply cybersecurity and data protection principles in their daily work to enable secure, compliant and resilient capabilities.	5	
11.10(j)	Controls for closed systems	The establishment of, and adherence to, written policies that hold individuals accountable and responsible for actions initiated under their electronic signatures, in order to deter record and signature falsification.	Functional	Intersects With	Rules of Behavior	HRS-05.1	Mechanisms exist to define acceptable and unacceptable rules of behavior for the use of technologies, including consequences for unacceptable behavior.	5	
11.10(j)	Controls for closed systems	The establishment of, and adherence to, written policies that hold individuals accountable and responsible for actions initiated under their electronic signatures, in order to deter record and signature falsification.	Functional	Intersects With	Technology Use Restrictions	HRS-05.3	Mechanisms exist to establish usage restrictions and implementation guidance for organizational technologies based on the potential to cause damage to Technology Assets, Applications and/or Services (TAAS), if used maliciously.	5	
11.10(k)	Controls for closed systems	Use of appropriate controls over systems documentation including:	Functional	Intersects With	Data Protection	DCH-01	Mechanisms exist to facilitate the implementation of data protection controls.	5	
11.10(k)	Controls for closed systems	Use of appropriate controls over systems documentation including:	Functional	Intersects With	Standardized Operating Procedures (SOP)	OPS-01.1	Mechanisms exist to identify and document Standardized Operating Procedures (SOP), or similar documentation, to enable the proper execution of day-to-day / assigned tasks.	5	
11.10(k)(1)	Controls for closed systems	Adequate controls over the distribution of, access to, and use of documentation for system operation and maintenance.	Functional	Intersects With	Disclosure of Information	DCH-03.1	Mechanisms exist to restrict the disclosure of sensitive and/or regulated data to authorized parties with a need to know.	8	
11.10(k)(2)	Controls for closed systems	Revision and change control procedures to maintain an audit trail that documents time-sequenced development and modification of systems documentation.	Functional	Subset Of	Configuration Change Control	CHG-02	Mechanisms exist to govern the technical configuration change control processes.	10	
11.30	Controls for open systems	Persons who use open systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, as appropriate, the confidentiality of electronic records from the point of their creation to the point of their receipt. Such procedures and controls shall include those identified in §11.10, as appropriate, and additional measures such as document encryption and use of appropriate digital signature standards to ensure, as necessary under the circumstances, record authenticity, integrity, and confidentiality.	Functional	Intersects With	Operationalizing Security, Compliance & Resilience Capabilities	GOV-15	Mechanisms exist to compel data and/or process owners to operationalize security, compliance and resilience practices for Technology Assets, Applications, Services and/or Data (TAASD) under their control.	8	
11.30	Controls for open systems	Persons who use open systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and, as appropriate, the confidentiality of electronic records from the point of their creation to the point of their receipt. Such procedures and controls shall include those identified in §11.10, as appropriate, and additional measures such as document encryption and use of appropriate digital signature standards to ensure, as necessary under the circumstances, record authenticity, integrity, and confidentiality.	Functional	Intersects With	Applied Security, Compliance and Resilience Controls Documentation	IAO-03	Mechanisms exist to generate authoritative documentation (e.g., System Security Plan (SSP)) that: (1) Identifies key architectural and implementation information on in-scope Technology Assets, Applications and/or Services (TAAS); (2) Reflects the current state of applied security, compliance and resilience controls on applicable People, Processes, Technologies, Data and/or Facilities (PPTDF) that are contained within the system boundary; and (3) Provides a historical record of applied security controls, including	8	
11.50	Signature manifestations	N/A	Functional	No Relationship	N/A	N/A	N/A	0	
11.50(a)	Signature manifestations	Signed electronic records shall contain information associated with the signing that clearly indicates all of the following:	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.	5	
11.50(a)(1)	Signature manifestations	The printed name of the signer;	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.	5	
11.50(a)(2)	Signature manifestations	The date and time when the signature was executed; and	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.	5	
11.50(a)(3)	Signature manifestations	The meaning (such as review, approval, responsibility, or authorship) associated with the signature.	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.	5	
11.50(b)	Signature manifestations	The items identified in paragraphs (a)(1), (a)(2), and (a)(3) of this section shall be subject to the same controls as for electronic records and shall be included as part of any human readable form of the electronic record (such as electronic display or printout).	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.	5	
11.70	Signature/record linking	Electronic signatures and handwritten signatures executed to electronic records shall be linked to their respective electronic records to ensure that the signatures cannot be excised, copied, or otherwise transferred to falsify an electronic record by ordinary means.	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.	5	
11.100	General requirements	N/A	Functional	No Relationship	N/A	N/A	N/A	0	
11.100(a)	General requirements	Each electronic signature shall be unique to one individual and shall not be reused by, or reassigned to, anyone else.	Functional	Subset Of	Identity & Access Management (IAM)	IAC-01	Mechanisms exist to facilitate the implementation of identification and access management controls.	10	

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
11.100(b)	General requirements	Before an organization establishes, assigns, certifies, or otherwise sanctions an individual's electronic signature, or any element of such electronic signature, the organization shall verify the identity of the individual.	Functional	Intersects With	Identity Proofing (Identity Verification)	IAC-28	Mechanisms exist to verify the identity of a user before issuing authenticators or modifying access permissions.	8	
11.100(c)	General requirements	Persons using electronic signatures shall, prior to or at the time of such use, certify to the agency that the electronic signatures in their system, used on or after August 20, 1997, are intended to be the legally binding equivalent of traditional handwritten signatures.	Functional	No Relationship	N/A	N/A	N/A	0	
11.100(c)(1)	General requirements	The certification shall be signed with a traditional handwritten signature and submitted in electronic or paper form. Information on where to submit the certification can be found on FDA's web page on Letters of NonRepudiation Agreement.	Functional	No Relationship	N/A	N/A	N/A	0	
11.100(c)(2)	General requirements	Persons using electronic signatures shall, upon agency request, provide additional certification or testimony that a specific electronic signature is the legally binding equivalent of the signer's handwritten signature.	Functional	No Relationship	N/A	N/A	N/A	0	
11.200	Electronic signature components and controls	N/A	Functional	No Relationship	N/A	N/A	N/A	0	
11.200(a)	Electronic signature components and controls	Electronic signatures that are not based upon biometrics shall:	Functional	No Relationship	N/A	N/A	N/A	0	
11.200(a)(1)	Electronic signature components and controls	Employ at least two distinct identification components such as an identification code and password.	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.	5	
11.200(a)(1)(i)	Electronic signature components and controls	When an individual executes a series of signings during a single, continuous period of controlled system access, the first signing shall be executed using all electronic signature components; subsequent signings shall be executed using at least one electronic signature component that is only executable by, and designed to be used only by, the individual.	Functional	No Relationship	N/A	N/A	N/A	0	
11.200(a)(1)(ii)	Electronic signature components and controls	When an individual executes one or more signings not performed during a single, continuous period of controlled system access, each signing shall be executed using all of the electronic signature components.	Functional	No Relationship	N/A	N/A	N/A	0	
11.200(a)(2)	Electronic signature components and controls	Be used only by their genuine owners; and	Functional	No Relationship	N/A	N/A	N/A	0	
11.200(a)(3)	Electronic signature components and controls	Be administered and executed to ensure that attempted use of an individual's electronic signature by anyone other than its genuine owner requires collaboration of two or more individuals.	Functional	No Relationship	N/A	N/A	N/A	0	
11.200(b)	Electronic signature components and controls	Electronic signatures based upon biometrics shall be designed to ensure that they cannot be used by anyone other than their genuine owners.	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.	5	
11.300	Controls for identification codes/passwords	Persons who use electronic signatures based upon use of identification codes in combination with passwords shall employ controls to ensure their security and integrity. Such controls shall include:	Functional	No Relationship	N/A	N/A	N/A	0	
11.300(a)	Controls for identification codes/passwords	Maintaining the uniqueness of each combined identification code and password, such that no two individuals have the same combination of identification code and password.	Functional	No Relationship	N/A	N/A	N/A	0	
11.300(b)	Controls for identification codes/passwords	Ensuring that identification code and password issuances are periodically checked, recalled, or revised (e.g., to cover such events as password signing).	Functional	No Relationship	N/A	N/A	N/A	0	
11.300(c)	Controls for identification codes/passwords	Following loss management procedures to electronically deauthorize lost, stolen, missing, or otherwise potentially compromised tokens, cards, and other devices that bear or generate identification code or password information, and to issue temporary or permanent replacements using suitable, rigorous controls.	Functional	Intersects With	Cryptographic Key Management	CRY-09	Mechanisms exist to facilitate cryptographic key management controls to protect the confidentiality, integrity and availability of keys.	5	
11.300(d)	Controls for identification codes/passwords	Use of transaction safeguards to prevent unauthorized use of passwords and/or identification codes, and to detect and report in an immediate and urgent manner any attempts at their unauthorized use to the system security unit, and, as appropriate, to organizational management.	Functional	Intersects With	Protection of Authenticators	IAC-10.5	Mechanisms exist to protect authenticators commensurate with the sensitivity of the information to which use of the authenticator permits access.	5	
11.300(e)	Controls for identification codes/passwords	Initial and periodic testing of devices, such as tokens or cards, that bear or generate identification code or password information to ensure that they function properly and have not been altered in an unauthorized manner.	Functional	Intersects With	Functional Review Of Security, Compliance & Resilience Controls	CPL-03.2	Mechanisms exist to regularly review Technology Assets, Applications and/or Services (TAAS) for adherence to the organization's security, compliance and/or resilience policies and standards.	5	